Namburi Hemanth Akhiland Kumar

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SKILLS

- **Languages**: C++, Python, SQL, JavaScript
- **AI/ML**: PyTorch, Torchvision, CNNs, ResNet18, transfer learning, scikit-learn, TF-IDF, Logistic Regression, spaCy, LLM integration (Llama 3.1 8B, Groq API), structured JSON outputs
- **Databases & Tools**: MySQL, MongoDB, Git, Linux
- **Backend & web:** Stream lit, Pydantic v2, asyncio, Node.js, Express.js, React.js, REST APIs, HTML, CSS
- **Core Concepts**: Data Structures & Algorithms, OOP, DBMS
- **Soft Skills**: Cognitive Agility, Solution-Oriented Mindset, Execution Focus, Learning Adaptability, Ownership Mentality

PROJECTS

ParkSense-AI | Parking Space Classification (CNN) Sep 2026 – Present

- Built a ResNet18 transfer-learning classifier in PyTorch that labels individual parking spaces as busy or free.
- Trained on 12,584 CNRPark image patches (80:20 train/validation split) and reached 94.4% validation accuracy 94.4% validation accuracy after the first epoch on CPU.
- Integrating the model into a parking application and extending training to five epochs (in progress).
- **Tech:** Python, PyTorch, Torchvision, ResNet18, CNRPark

Intelligent E-Governance Grievance Center | LLM Application | Link May 2026 – Jul 2026

- Built an LLM pipeline that turns free-text citizen grievances into structured analyses: department routing, severity, priority score, sentiment, escalation risk and SLA assignment.
- Implemented Pydantic v2 schema validation and workflow verification checks to ensure reliable, structured grievance data.
- Built an asynchronous processing pipeline and Streamlit dashboard for complaint review and automated response letters, while profiling latency and throughput and achieving 90.7% validation accuracy
- **Tech:** Python, Groq API, Llama 3.1 8B, Streamlit, Pydantic v2, asyncio, Git, GitHub

Portfolio Optimization Using Genetic Algorithm | Optimization & Machine Learning | Link Oct 2025 – Dec 2025

- Implemented a genetic algorithm (selection, crossover, mutation, fitness evaluation) to search for the asset mix that best balances expected return and risk.
- Quantified portfolio returns and risk with NumPy and Pandas and used the results to rank candidate allocations and recommend an optimal portfolio.
- Analyzed datasets and created optimal investment recommendations by quantifying risk and return using various methods and performing a heuristic search to find the optimal solution.
- **Tech:** Python, Genetic Algorithm, NumPy, Pandas, Financial Analytics

TRAINING

Summer Training June 2025 – July 2025

Data Structures and Algorithms | Link

- Achievements: Developed an AI-powered Crop Disease Prediction System, which includes interactive visualization, confident predictions and weather-assisted disease analysis using HTML, CSS and JavaScript for the front-end.
- Key Learnings: Core Data Structures & Algorithms have been put to good use to optimize and facilitate data processing, searching and in general the application's overall performance, and to improve problem solving skills.
- Technical Exposure: In-depth working in C++, HTML, CSS, JavaScript, Git, and Algorithm designing in the assignments and the implementation of the project.

CERTIFICATES

- Oracle Agentic AI Certified Foundations Associate | Oracle — Sep 2026
- The Bits and Bytes of Computer Networking | Google (Coursera) — Sep 2024

ACHIEVEMENTS

- Solved 104 algorithmic problems on LeetCode across three difficulty levels, strengthening proficiency in data structures, algorithms, and computational problem-solving| Profile

EDUCATION

- **Lovely Professional University |** August 2023 – 2027
  Bachelor of Technology in Computer Science and Engineering; CGPA: 6.61 — Punjab, India
- **Narayana Junior college |** August 2022 – 2023
  Higher Secondary; Percentage: 83.3% — Andhra Pradesh, India
- **Lotus High School |** August 2020 – 2021
  Matriculation; Percentage: 99% — Andhra Pradesh, India